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Paleomagnetism of Palawan, Philippines

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Abstract

Paleomagnetic studies have been carried out on Palawan and on the island of Busuanga to the north. Results from the Cretaceous Espina Basalts of the Calatuigas Ophiolite in the South Palawan Block (SPB) pass a fold test, yield normal and reversed directions with a magnetic intensity and AF demagnetization characteristics consistent with a primary TRM. The mean direction is 293.9° and an inclination of 5.8°, with a k of 37.7 and an α_{95} of 12.6°. This suggests that these ophiolites have moved northward and rotated counterclockwise by $66° \pm +13°$ with respect to the geocentric axial dipole (GAD) field. It also suggests that they were obducted from the south.

Paleomagnetic directions from the Jurassic Busuanga Cherts and the Cretaceous Guinlo Formation from the island of Busuanga in the North Palawan Block (NPB) and from the Guinlo on the main island of Palawan are similar, fail regional fold tests, and have AF demagnetization characteristics consistent with secondary magnetization. Their inclinations are statistically indistinguishable at a 95% significance level, but variation in declination suggests differential local rotation about a vertical axis. The paleolatitude is comparable to that of regions of pervasive Cretaceous remagnetization in the South China borderland and may reflect similar remagnetization, consistent with the NPB's proposed South China origin. © 2000 Elsevier Science Ltd. All rights reserved.

1. Introduction

Palawan is located to the west of the main Philippine archipelago (Fig. 1). Its northeast–southwest orientation contrasts with the sinusoidal, generally north-south orientation of the remainder of the Philippine Archipelago. Palawan is made up of more than a thousand islands and islets. The South China Sea to the north and the Sulu Sea to the east isolate the province from the rest of the Philippines. The main elongate island is about 450 km long. It narrows from its maximum width of approximately 50 km near the Municipality of Brooke's Point to about 9 km north of Puerto Princessa.

The geology of Palawan is described in detail in the first volume of Geology and Mineral Resources of the Philippines by Fernandez (1981) and a more recent compilation is given by Almasco (1994). A brief overview is given here. Fig. 2 is a geological map of Northern Palawan and summarizes its stratigraphy. Fig. 3 is a similar map for Southern Palawan. A major north–south fault at the narrowest part of the island north of Puerto Princessa defines the boundary between the South Palawan block (SPB) from the North Palawan Block (NPB).

In the NPB, unmetamorphosed Paleozoic shales and limestones of the Permian Bacuit formation lie unconformably on the Permo-Carboniferous metamorphics. The Mesozoic section begins with the recrystallized limestone of the Minilog formation, which is followed by the radiolarian cherts of the Middle Triassic Limangcong and the Jurassic Busuango formations and the Cretaceous Guinlo formation. The latter was intruded by the late Eocene to early Oligocene Kapoas Granites. A complex body of serpentine and gabbroic

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rocks of Late Cretaceous age is found in the southernmost part of the NPB. The Late Eocene to Middle Miocene marine sequence is largely limestone and shale, but includes pillow lavas and an andesite. The section is completed by the Pleistocene Manguao volcanics and Quaternary alluvium.

The SPB stratigraphy is quite different from the section in the NPB with a total absence of unmetamorphosed Paleozoic and Mesozoic sediments. Instead there are Eocene schists and metavolcanics. However, the Stavely Range gabbros are equivalent to the complex serpentines and gabbroic rocks of the NPB. The chert-spilite group found in the SPB is equivalent to the Espina Formation in Balabac Island which lies to the south between Palawan and Northern Borneo. Raschka et al. (1985) recognized these sequences as an ophiolite, lacking only a sheeted dyke complex.

Hashimoto and Sato (1973) discussed the structural relationship between South and North Palawan, reporting that the two regions are separated by a zone of left-lateral strike-slip faulting along the fault north of Puerto Princessa. The faulting is manifested not by a single break, but by numerous minor faults in serpentinite and gabbro units. Hashimoto and Sato further noted that the metamorphic belt in Southern Palawan trends northeast–southwest, parallel to the elongation axis of the island, whereas the gross structure of North Palawan is oblique to it. They postulated that the region was affected not only by pre-Middle Jurassic deformation and metamorphism, but also by later deformations that are represented by kink bands and strike-slip faulting such as that in the Baheli Isthmus area, north of Puerto Princessa. They further noted that the sedimentary units of probable Paleogene age that cover the crystalline and basic rocks in the Baheli Isthmus area are also folded, which may indicate Tertiary deformation.

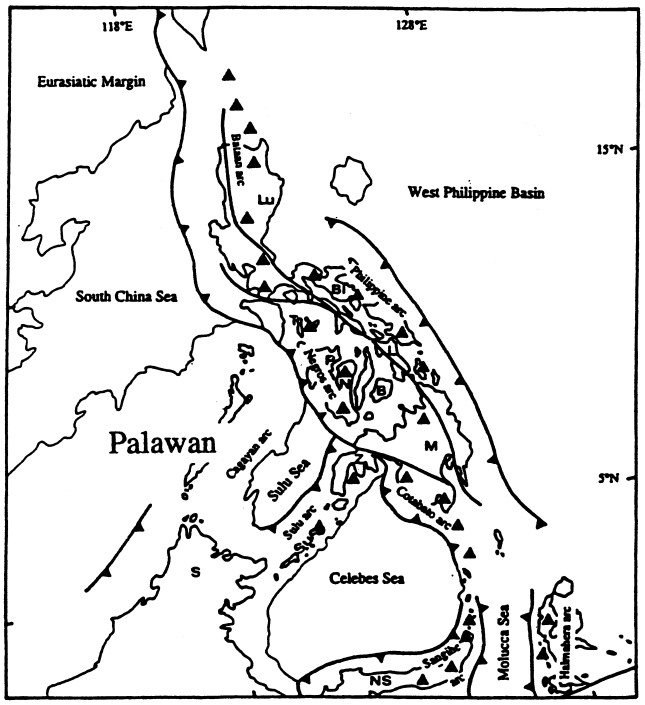


Fig. 1. Simplified regional tectonic map giving location of Palawan. Quarternary Arc Volcanism (triangles): Bohol (B), Bicol (Bi), Leyte (L), Luzon (Lu), Mindanao (M), Negros (N), North ARM of Sulawesi (NS), Panay (P), Sabah (S), Sulu Islands (Su), Tablas (T) and Zamboanga (Z) (after Cambray et al., 1995).

The contrast between the NPB with its biostratigraphically dated Paleozoic and Mesozoic sections (Hashimoto and Sato, 1973) and the absence of unmetamorphosed Paleozoic sedimentary rocks in the SPB led to the suggestion of two different terranes (Tamesis et al., 1973; Hamilton, 1979; Taylor and Hayes, 1980, 1983; Holloway, 1982; McCabe et al., 1982; Karig, 1983). The NPB was interpreted to be a rifted fragment from China (Holloway, 1982). The Southern Palawan Block (SPB) with its late Mesozoic and Cenozoic ophiolites and a late Paleocene–early middle Miocene clastic sequence that may be correlatable with the Crocker Formation of Borneo was thought to have a different origin. However, Hinz and Schluter (1985) carried out geophysical exploration in Dangerous Grounds, South China Sea and the margin of SW Palawan and recognized an Upper Oligocene to Lower Miocene carbonate platform with a sequence of regional unconformities, which they correlated with events in the history of rifting and seafloor spreading in the South China Sea. They suggested that this region is underlain by stretched continental crust and that both the NPB and the SPB are parts of a microcontinent carried south by South China seafloor spreading.

In this paper, we describe a paleomagnetic study of Palawan undertaken to determine the paleomagnetic signatures of the North and South Palawan Blocks.

2. Previous paleomagnetic studies in the vicinity of Palawan

The paleomagnetism of the Philippines has been reviewed recently (Fuller et al., 1991; Van der Voo, 1993) and its interpretation remains problematical, which is perhaps not surprising given its complex history of multiple subduction events, collisions and strike slip faulting. However, there is agreement (e.g. Hsu 1972; Fuller et al., 1983, 1991; McCabe et al., 1987) that there is no reliable paleomagnetic evidence for significant rotations, or translations, in the Plio-Pleistocene. In contrast, Late Miocene sites in Zambales and the Central Cordillera of Luzon show CW rotations. These CW rotations were interpreted as evidence of a regional rotation of Luzon, when it was

attached to the Philippine Sea Plate prior to the development of the East Luzon trough. Other Late Miocene results from the central and southern part of the Philippine Arc show no rotation.

From six older Cenozoic sites in Panay Island, McCabe et al. (1982) demonstrated consistent clockwise rotations, in contrast to the counterclockwise rotation of early Miocene and older sites in Marinduque and Luzon to the north. They suggested that the Panay result was evidence of the collision of the Philippine Arc with the Northern Palawan Block. Paleomagnetic results from Jurassic sections in Mindoro yielded similar paleolatitudes to Jurassic sites in South China, which was recognized to be consistent with the

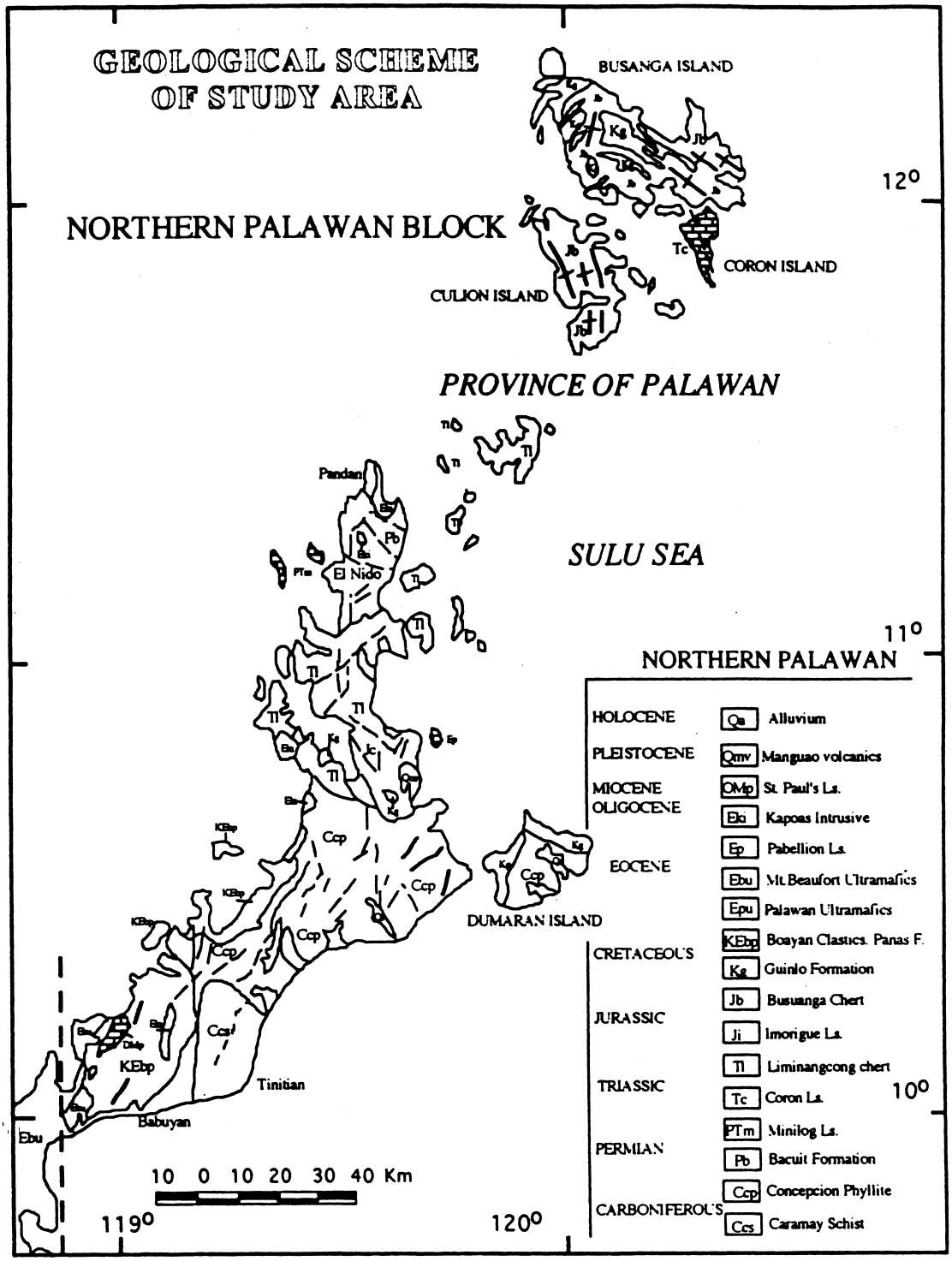


Fig. 2. Geological map of North Palawan Block (Almasco, 1994).

hypothesis that the North Palawan Block rifted away from Southern China (Holloway, 1982, McCabe et al., 1987). The rotation of the Mindoro VGP from the South China pole was interpreted to be due to local shears at the edge of the North Palawan block.

In 1989, Leg 124 of the Ocean Drilling Program (ODP) investigated the Sulu Sea and Celebes Sea marginal basins. Paleomagnetic measurements were conducted on the cores to evaluate the relationship of these two basins with the West Philippine Basin, North Indian Ocean and Molucca Sea. Shibuya et al. (1991) concluded from their paleomagnetic measurements of Celebes Sea cores that this basin was rotated 60° CCW in Oligocene time, and that its paleolatitude remained low throughout its history. Results from the

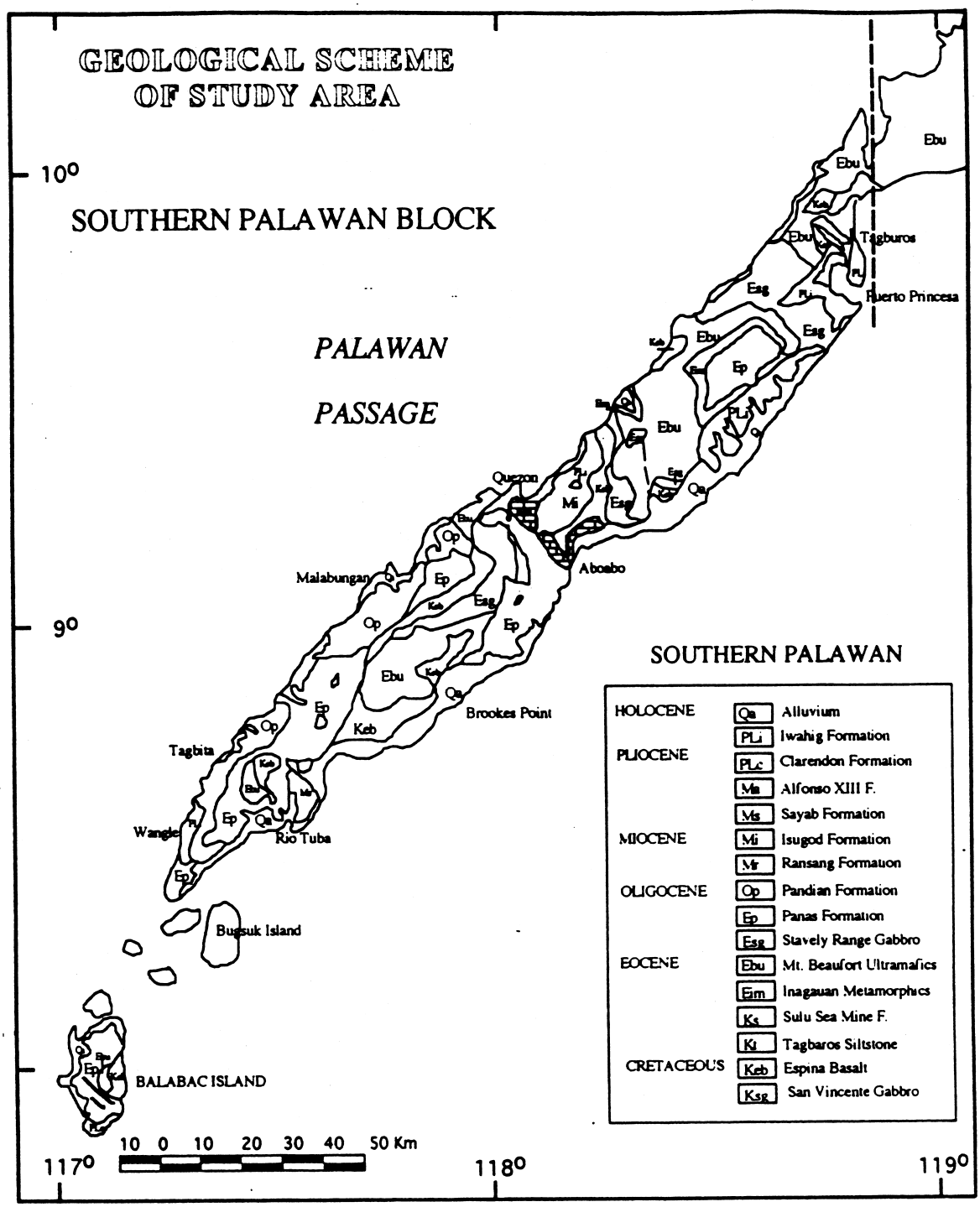


Fig. 3. Geological map of South Palawan Block (Almasco, 1994).

Sulu Sea by Hsu et al. (1989) suggest that the Cagayan Ridge has neither rotated, nor migrated latitudinally, since early middle Miocene times.

Paleomagnetic studies in the Philippines, in the Sulu and Celebes Seas, in Borneo and Peninsular Malaya form elements of a framework within which to interpret the results from Palawan. The Eurasian Apparent Polar Wander (APW) path (Besse and Courtillot, 1991) gives a reference data set with which to compare all of these results from SE Asia with those for stable Eurasia and similarly the South China APW path (Enkin et al., 1991) permits comparison with that landmass.

3. Paleomagnetic survey of Palawan

The paleomagnetic results obtained in Palawan are discussed in terms of the North and South Palawan Blocks. A total of 147 paleomagnetic sites were collected in the region. Only 38 sites were accepted as giving consistent data, using a criterion of an $\alpha_{95} < 20°$,

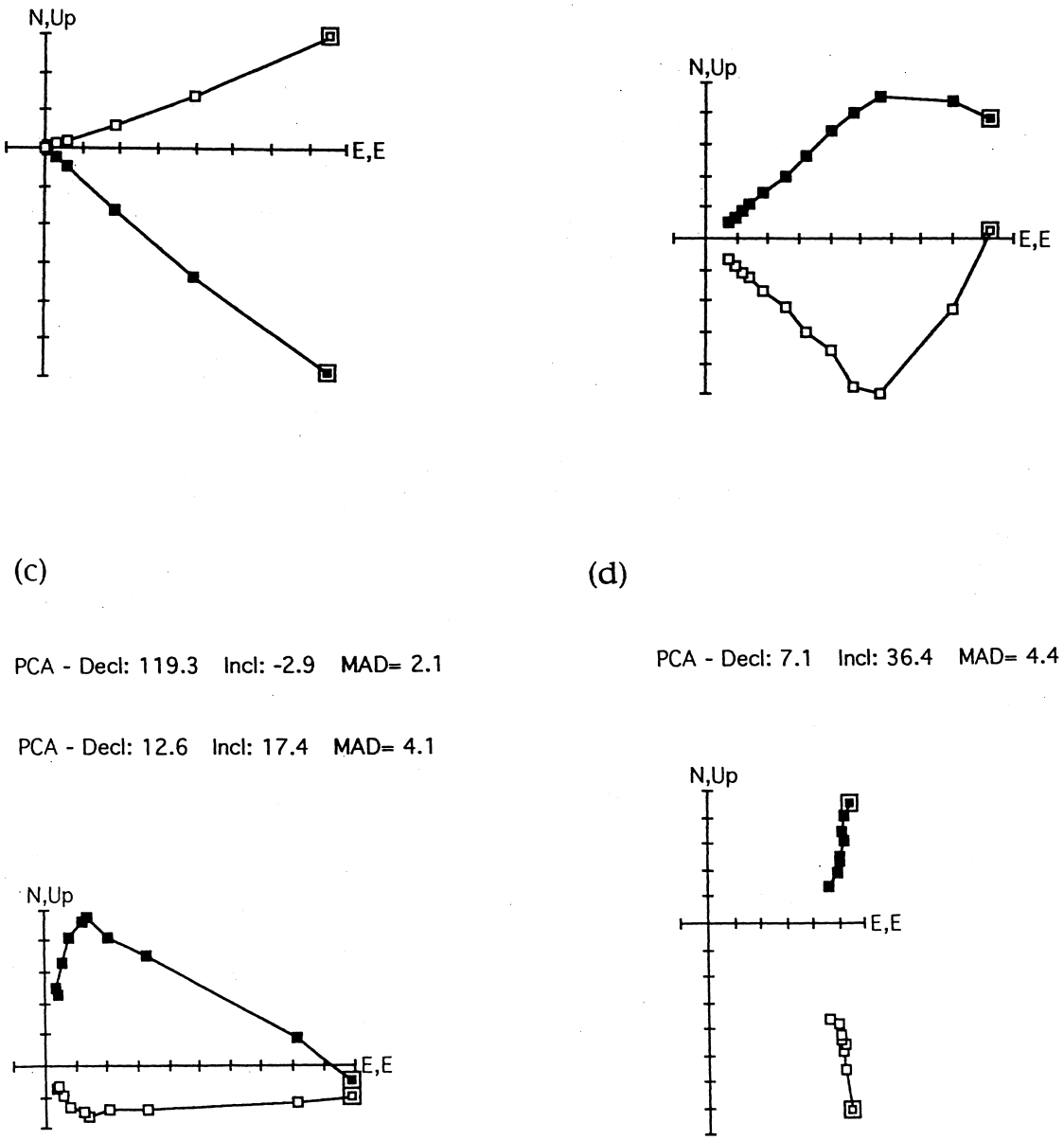


Fig. 4. Representative demagnetization plots. (A) Essentially single direction of magnetization converging to origin throughout most of demagnetization. MAD angle is less than 1. (B) Mixture of soft and hard magnetizations which are largely separable and the hard magnetization convergent to origin with MAD angle less than 2. (C) Mixtures of magnetizations which are partially separable to yield 2 or more directions with MAD angles between 2 and 5. (D) Mixtures of magnetizations which are not well separated by demagnetization. This group includes samples which do not demagnetize, or give erratic results, although a soft near present field directions may be poorly resolved.

although data from many of these sites were of a higher quality, yielding a far smaller α_{95}.

3.1. Paleomagnetic methods

Standard collection and analysis methods were used throughout the study. About eight samples were collected per site with a gasoline powered drill. In some cases, when the rock was too friable, too fractured, or too hard to drill satisfactorily in the field, hand samples were collected and cores drilled from them in the laboratory. The measurements of magnetization were made on a 2G^{TM} RF SQUID superconducting magnetometer and with a Molspin^{TM} spinner magnetometer. Fisher (1953) statistics and Kirschvink (1980) principal component analysis were utilized.

AF demagnetization was carried out with a 2G^{TM} demagnetizer. Stepwise and continuous systems developed at University of California at Santa Barbara (e.g. Dunn and Fuller, 1984 and Fuller, 1987) were used for thermal demagnetization. In presenting the Zijderveld data (Zijderveld 1967), we have adopted the procedure of classifying the results into subsets and giving their representation at the various sites. The Maximum Angular Dispersion (MAD) value used in the classification is the maximum departure from the mean direction brought about by varying each of the three components by one standard deviation. The analyses are always unanchored, that is to say they are not constrained to pass through the origin. Examples of the various subsets are shown in Fig. 4. Type A plots consist of essentially a single direction of magnetization giving near to straight-line convergence to the origin with MAD less than 1°. Type B has more than one direction recorded, one of which is a soft magnetization removed at low fields, leaving near to straight-line convergence to the origin with a MAD of less than 2°. Type C has more than one direction of magnetization, but none is very soft, although some separation of moments is possible yielding straight-line segments with MADs of less than 2.5°. Type D curves do not show a well defined straight-line segment convergent to the origin from which an isolated direction can be determined, although a soft secondary magnetization may be poorly resolvable ($2° < MAD < 5°$).

Samples were subjected to systematic rock magnetism analysis in the form of (1) low field susceptibility measurements as a function of temperature to identify magnetic phases present, (2) acquisition and demagnetization of IRM (Isothermal Remanent Magnetization) were determined to construct Cisowski diagrams (Cisowski, 1981), and (3) comparison of NRM characteristics with IRMs (e.g. Fuller et al., 1988).

The low field susceptibility as a function of temperature was determined using a ballistic method. A PAR phase sensitive detector was used to monitor the output of a pair of balanced pick up coils within the main drive coil which gave an AC field of 1 mT (J. R. Dunn C. and Anderson, private communication 1991). Fig. 5a shows an example of the results. In this case, the Curie point is in the vicinity of 550 and 600°C. In other examples different Curie temperatures were found and in some cases the observations were consistent with the formation of magnetite during heating.

The Cisowski diagrams combine information about carriers, domain state and interactions between magnetic phases. The example shown is a typical result consistent with the presence of fine, single domain, or pseudosingle domain, magnetite, which is non-interacting. The symmetry of the curves with the cross over value (R) close to 0.5 indicates minimal interactions and the remanent coercivity (Hrc) considered in conjunction with independent phase identification of titanomagnetite indicate fine particles (e.g. Day et al., 1976, 1977). Multidomain carriers give much softer demagnetization and acquisition IRM curves with Hrc close to 10 mT.

The comparison of the magnitude and demagnetization characteristics of the NRM with those of saturation IRM permits, in some instances, a distinction between primary and secondary magnetizations. For example, in rocks carrying TRM (Thermal Remanent Magnetization), the ratio of NRM:IRMs is of the order of 10^{-2}, so that smaller values imply secondary magnetization. In the example shown in the left hand illustration, a typical TRM-like curve is seen with a ratio of NRM:IRMs of the order of 10^{-2}. In contrast in the second example the ratio is much lower indicating that it is not carrying a primary TRM. In each case a soft magnetization is demagnetized before the main diagnostic curve is reached. In sedimentary rocks carrying a primary DRM (Depositional Remanent Magnetization), the ratio of NRM:IRMs is of the order of 10^{-3}, so that larger values, indicate a more efficient process such as chemical remanence which will often imply secondary magnetizations. Again, curves for DRM and TRM in comparison with IRMs can sometimes help in interpreting the primary, or secondary nature of the NRM.

If possible, fold tests were performed, but for the most part a regional attitude test was all that could be done. In this, local bedding is used to correct to stratigraphic coordinates for the comparison with in situ directions.

3.2. Site descriptions and results

3.2.1. North Palawan Block

Collections were made from the Paleozoic to Mesozoic rocks in the northeastern part of Palawan Island and in Busuanga Island. The major lithological units studied were pelagic cherts, carbonates, and flysch-type

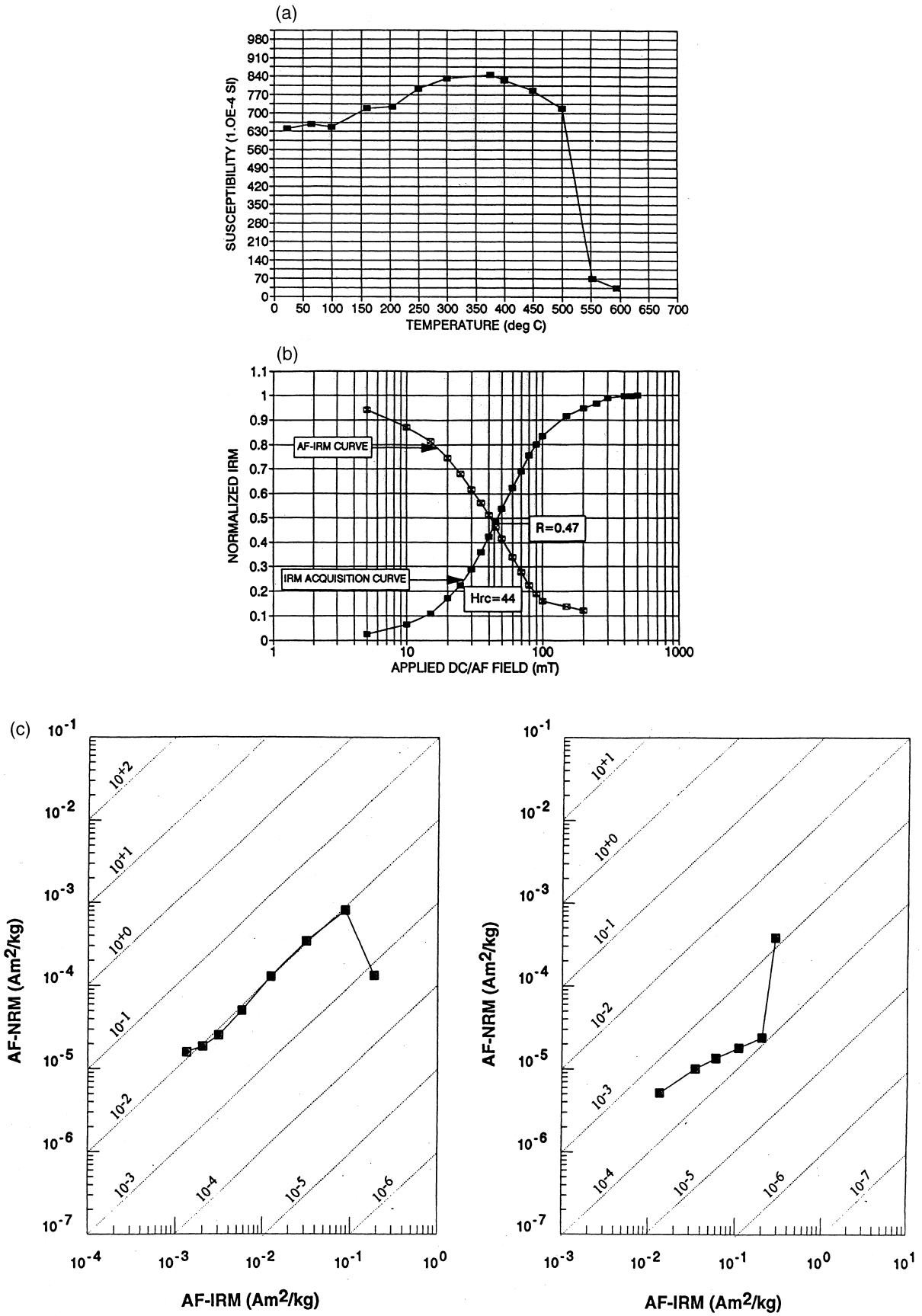


Fig. 5. Examples of rock magnetic analysis: (a) thermal dependence of weak field susceptibility to identify Curie points, (b) acquisition and demagnetization of IRMs, and (c) AF demagnetization characteristics of NRM and IRMs.

turbidites. Hashimoto and Sato (1973) called these assemblages the Malampaya Sound Group which included Bacuit, Minilog, Liminangcong, and Guinlo Formations. Middle-late Permian, Early Jurassic and Late Jurassic fossils were observed, in addition to the Middle Triassic fossils identified by Fontaine (1979) and Wolfart et al. (1986). The results from the Busuanga Island are described first followed by those from the main island of Palawan, which are separated into those from the NPB and SPB. In each case, the results are described in age order of the formations from younger to older.

3.2.1.1. The Cretaceous Guinlo Formation (King Ranch Formation) from Busuanga Island (North Palawan Block). Sites Bu 3, 19, 20, 21, and 22 were drilled in the turbiditic sandstone of the King Ranch Formation (Fig. 1) and yielded interpretable paleomagnetic results. The sites are shown in Fig. 6 and the results in Fig. 7 and Table 1.

NRM. Both alternating-field and thermal demagnetization were used to clean the specimens, but the AF method tended to leave a substantial moment even after demagnetization to 100 mT. Zijderveld plots for thermal demagnetization all reveal linear segments with a majority also showing type A, or B plots. The harder direction (Fig. 7) has a mean declination of 32.8° and inclination of 44.1° with an α_{95} of 5.2° and k of 216. This direction failed a regional tilt test with k falling from 216 to 6 and α_{95} increasing from 5.2° to 33.8°. There is a significant spread in the in situ direction with roughly constant inclination such that the declination varies from 22.9° to 39.4°.

Rock magnetism. The thermal dependence of weak field susceptibility of samples from all sites revealed Curie points consistent with presence of magnetite, although samples from sites Bu 3 and Bu 20 had such large increases above 350°C that it appeared that magnetite was being formed during the heating process. The acquisition of IRM and its demagnetization, with

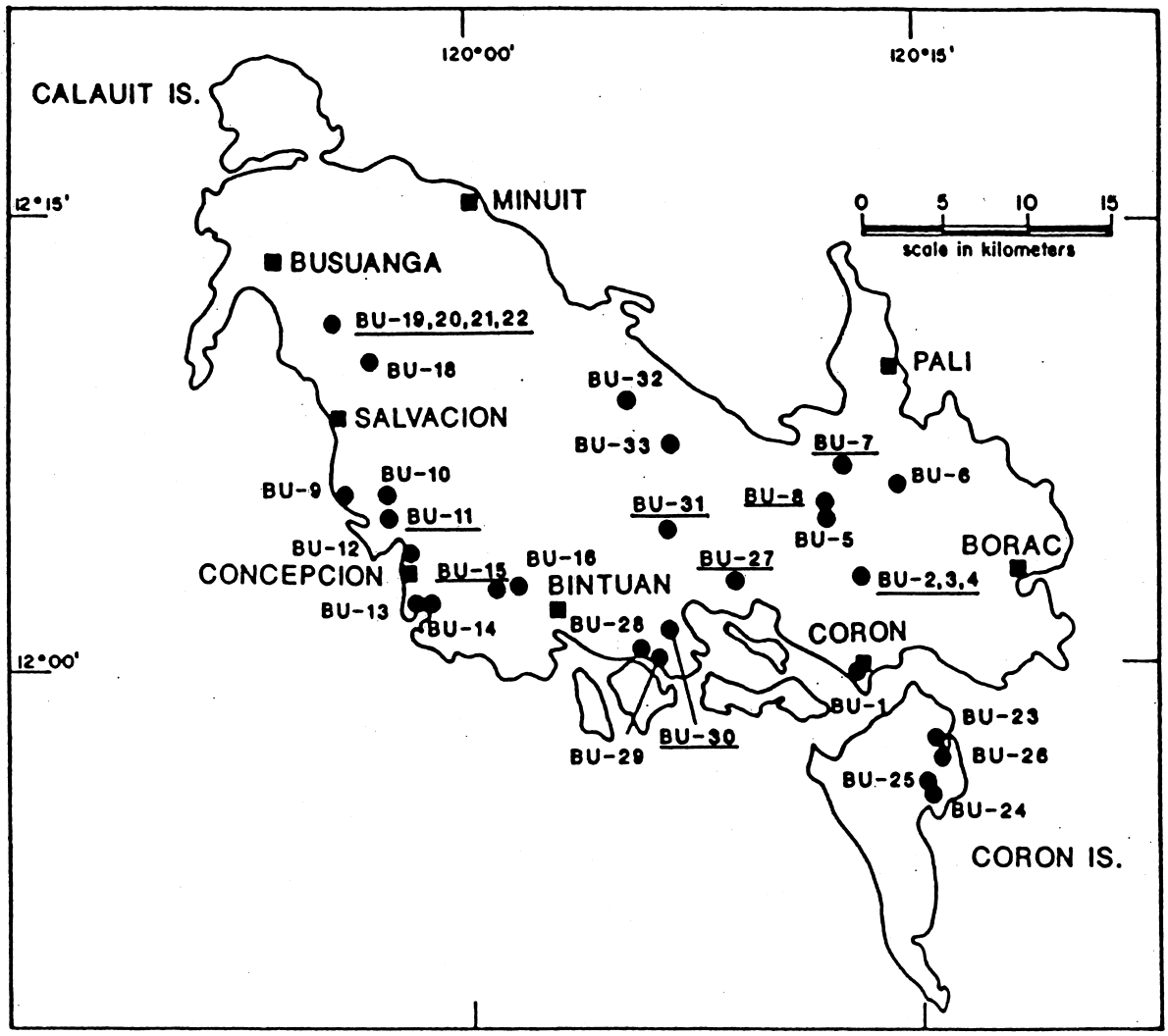


Fig. 6. Site location map for Busuanga Island.

R values near to 0.5 and Hrc around 50 mT, indicate the presence of fine grain non-interacting magnetite in sample at all sites. At site Bu 22 the non-saturating IRMs acquisition suggests the presence of some hematite. The ratio of NRM:IRMs is too high for the remanence to be a simple DRM or PDRM.

Interpretation. It appears from the attitude test and the NRM:IRMs ratio that the King Ranch formation has a secondary magnetization acquired after the rocks achieved their present attitude and that it is predominantly carried by fine grained magnetite. This magnetization was clearly not acquired in the present field. The wide scatter of directions suggests differential local rotation about a vertical axis.

3.2.1.2. The Jurassic Busuanga and Limangong Formations from Busuanga Island (North Palawan Block). The four sites in the Busuanga Formation from which we obtained interpretable results were BU 7, 8, 15 and 27 and are shown in Fig. 6. The results are given in Fig. 8 and Table 1. Three sites in the older Limangong Formation are also indicated in Fig. 6 and the results given in Table 1.

NRM. Alternating-field and thermal demagnetization were both effective in cleaning the Busuanga chert specimens magnetically. However, the alternating-field method removed a low inclination low coercivity magnetization more efficiently. This was removed by 15 mT and was followed by the isolation of a single direction, as shown by convergence to the origin in type B Zijderveld plots. The majority of the samples showed this behavior, although in about 10% good convergence to the origin was not observed resulting in type C plots in which linear segments are evident, but MAD values have increased. The directions found yielded a mean declination of 60.5° and an inclination of 44.9°, with an α_{95} of 17.3° and a k of 29.1 (Fig. 8). This direction failed the tilt test, yielding an α_{95} of 49.7° and a k of 4.4 after correction to stratigraphic coordinates.

Site BU 11 in the Limangong Formation gave a present field direction, whereas BU 30 and 31 yielded directions similar to those for the Busuanga Chert in geographic coordinates.

Rock magnetism. Observations of weak field susceptibility as a function of temperature yielded curves with broad Hopkinson rises and Curie points in the vicinity of 550°C, suggesting the presence of fine grained magnetite as the principal magnetic phase. IRMs acquisition and demagnetization was also consistent with the presence of fine magnetite, although in some samples there appeared to be significant amounts of a higher coercivity phase. Plots of AF-demagnetization characteristics gave NRM:IRMs ratios too large to be depositional remanence, but consistent with a secondary chemical remanence in these cherts.

Interpretation. The observations suggest that the Busuanga Cherts of the North Palawan group have a secondary magnetization carried predominantly by fine magnetite. This is an ancient secondary magnetization. The results from the Limangong yielded one present field direction and two sites with directions similar to those found in the Busuanga cherts. However, because only 2 sites yielded useful information the results from this formation have not been included in our analysis.

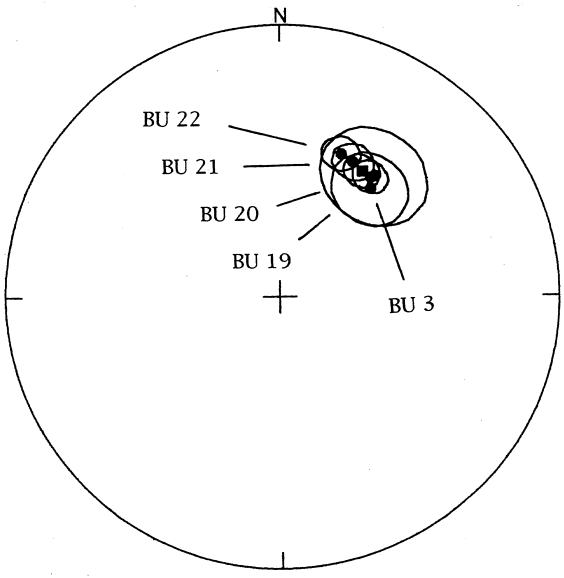


Fig. 7. Results from the King Ranch Formation (equivalent to Guinlo formation) from sites Bu 3, 19, 20, 21, and 22 from Busuanga Island. The data are plotted in situ and reflect a secondary magnetization acquired after the rocks acquired their present attitude.

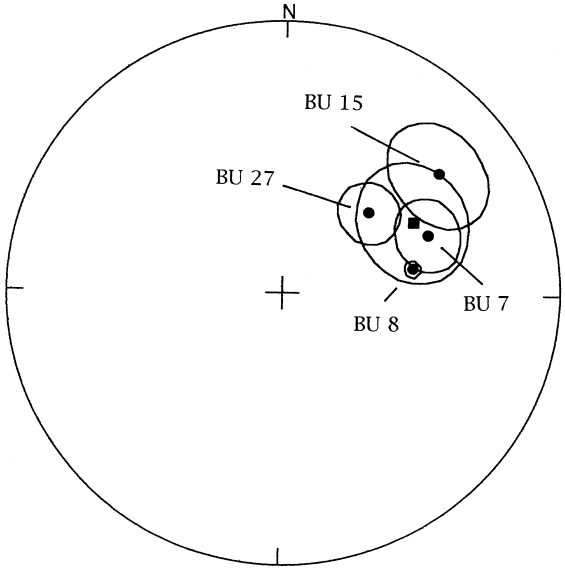


Fig. 8. Results from sites Bu 7, 8, 15 and 27 in the Busuanga chert of Busuanga Island. The data are plotted in situ and reflect a secondary magnetization acquired after the rocks acquired their present attitude.

Table 1
Paleomagnetic results from the North Palawan Block including Busuanga Island and Northern Palawan Island

Site No.	Age	Formation	Lithology	Latitude	Longitude	N	Geographic mean				Bedding correction	Stratigraphic mean			
							Declination (deg)	Inclination (deg)	α_{95} (deg)	k		Declination (deg)	Inclination (deg)	α_{95} (deg)	k
Busuanga Island															
BU-3	Cretaceous	King FM	Shale	12.0N	120.2E	5	39.4	47.1	11.1	48.7	330 39NE	46.2	9.8	11.1	48.8
BU-19	Cretaceous	King FM	Sandstone	12.2N	119.9E	5	37.7	42.9	15.2	26.3	189 18NW	20	48.9	15.3	26
BU-20	Cretaceous	King FM	Sandstone	12.2N	119.9E	8	36.5	45	5.1	120.2	230 84SE	95	13.6	5	122.4
BU-21	Cretaceous	King FM	Sandstone	12.2N	119.9E	9	28.2	42.7	5.8	79.2	230 84SE	94.2	20.8	6	74.7
BU-22	Cretaceous	King FM	Sandstone	12.2N	119.9E	5	22.9	42	5.4	202.5	235 80SE	95.3	24.5	5.2	214.7
		Mean				5	32.8	44.1	5.2	216	Variable	74.2	26.4	33.8	6.1
BU-7	Jurassic	Busuanga Ch	Chert	12.1N	120.2E	4	67.1	42.5	10.4	79	205 39NW	21	56.1	10.4	79.3
BU-8	Jurassic	Busuanga Ch	Chert	12.1N	120.2E	4	78.5	50.4	2.5	1403.6	313 70NE	67.8	−11.4	6.5	201.1
BU-15	Jurassic	Busuanga Ch	Chert	12.0N	120.0E	5	51.4	29.3	15.2	26.2	Variable	84.4	41.4	25.1	10.2
BU-27	Jurassic	Busuanga Ch	Chert	12.0N	120.2E	4	46	54.6	9.2	100.8	Variable	22.3	25.7	17.3	29.1
		Mean				4	60.5	44.9	17.3	29.1	Variable	50.7	31.3	49.4	4.4
BU-11	Permo/TR	Limangong	Chert	12.1N	119.9E	8	4.1	20	2.8	380.8	341 34NE	10.9	4.4	2.8	386.2
BU-30	Permo/TR	Limangong	Chert	12.0N	120.1E	4	65.8	68.6	21.9	18.6	295 78NE	38.9	−4.4	21.9	18.6
BU-31	Permo/TR	Limangong	Sandstone	12.1N	120.1E	4	50.7	35.6	30.4	10.1	146 54NE	52.3	−17.5	30.3	10.1
Northern Palawan Island															
CAL-1	Pleistocene	Manguao	Basalt flow	10.7N	119.6E	5	10.2	12.1	9.4	67.6	0	10.2	12.1	9.4	67.6
CAL-2	Pleistocene	Manguao	Basalt flow	10.7N	119.6E	4	13	16.7	7.5	149.4	0	13	16.7	7.5	149.4
T-5	Cretaceous	Guinlo FM	Sandstone	10.8N	119.4E	4	75.4	43.5	10.8	73.8	10 65SE	81.8	−17.4	11	71.1
T-6	Cretaceous	Guinlo FM	Sandstone	10.8N	119.4E	5	78.4	40.9	6.3	150.1	10 65SE	83	−20.8	6.3	147
PAL90-5	Permian	Bacuit	Shale	11.2N	119.4E	6	0.5	30	8.4	64.1	160 57SW	322.8	31.5	8.6	61.3
PAL90-6	Permian	Bacuit	Shale	11.2N	119.4E	6	353.8	40.2	6.7	101.4	352 55SW	318.6	22.3	6.7	101.5
PAL90-7	Permian	Bacuit	Limestone	11.2N	119.3E	6	1.7	36.5	11.6	34.3	Variable	353.8	40.2	6.7	34.3
PAL90-8	Permian	Bacuit	Shale	11.2N	119.4E	8	356.2	27.6	11.3	29.3	156 57SW	321.6	29.8	11.4	29.2

3.2.1.3. Manguao volcanics from North Palawan Island. The Manguao volcanics is a Pleistocene formation made up of basalt flows rich in olivine. The sites are shown in Fig. 9 and the results given in Table 1 and Fig. 10.

NRM. Sites (CAL-1 and CAL-2) taken from this formation were demagnetized solely by AF, which yielded predominantly excellent single direction behavior with type A curves. Some specimens showed two magnetic components with type B curves, but converged on the origin to yield the same direction as did the other samples.

The mean direction from this formation (Fig. 10) gave a declination of 11.1° and inclination of 14.4° with an α_{95} of 11.2° and k of 501. The VGP that corresponds to this mean direction has a longitude of 226.3°E and latitude of 78.5°N.

Rock magnetism. The temperature dependence of the weak field susceptibility of these samples shows a sharp decrease above 500°C with a Curie Point between 550 and 580°C indicating the presence of a low titanium titanomagnetite, or magnetite. AF demagnetization characteristics indicated the magnetite was fine single domain, or pseudosingle domain.

Interpretation. The departure from the geocentric axial dipole (GAD) field direction is within the range of variability to be expected from secular variation and we interpret this as a record of the Pleistocene geomagnetic field.

3.2.1.4. The Cretaceous Guinlo Formation from Northern Palawan Island. Sites T5 and T6 in the Guinlo on the main island of Palawan are shown in Fig. 9 and the results given in Table 1 and Fig. 11.

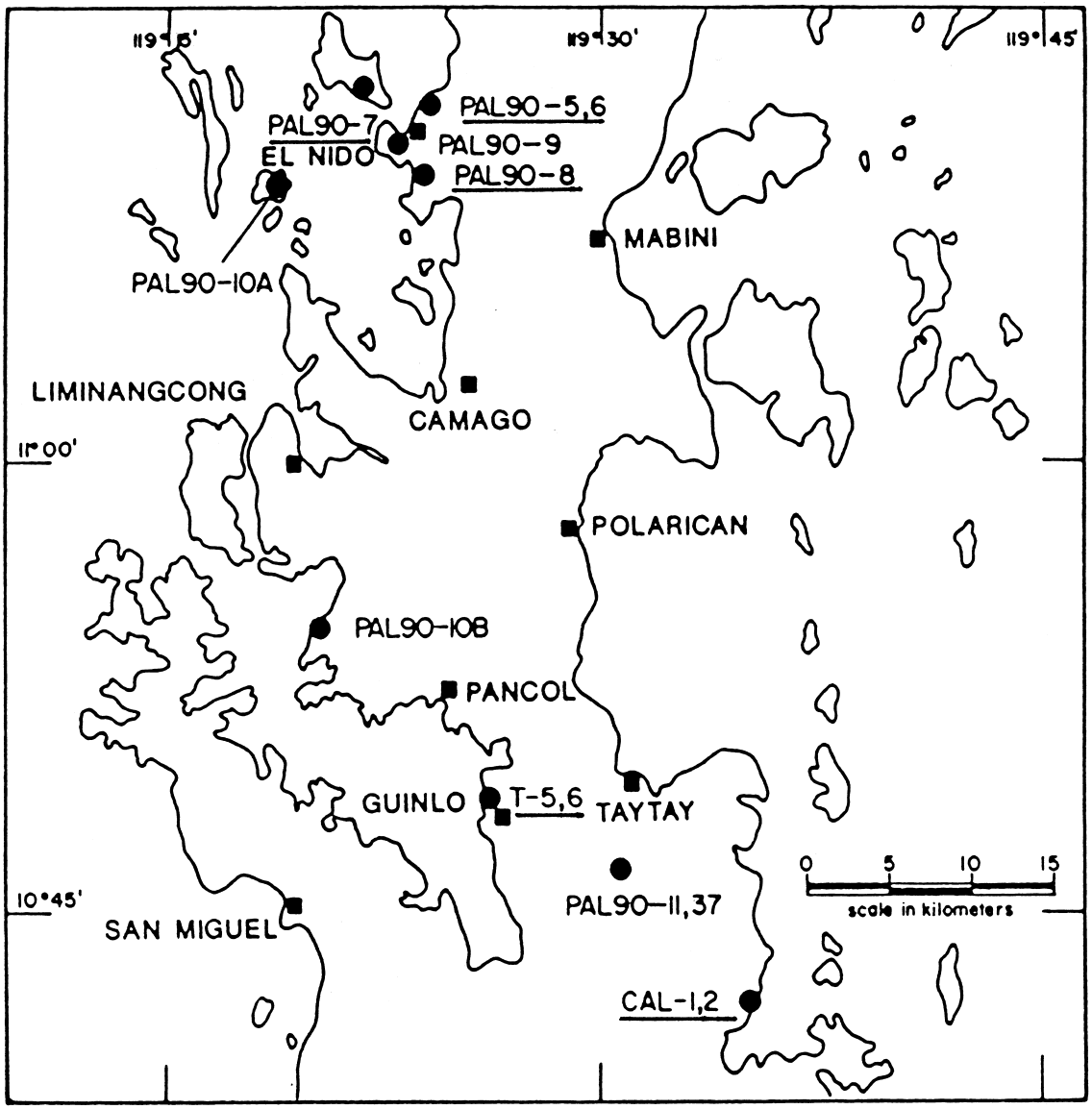


Fig. 9. Site location map for Northern Palawan.

NRM. AF demagnetization failed to demagnetize the samples, but thermal demagnetization yielded curves indicative of two phases, one unblocking in the vicinity of 300°C and the other having distributed blocking temperatures up to about 550°C. However, both phases carried similar directions of magnetization. The AF demagnetization Zijderveld plots were of type D, but for thermal demagnetization plots were predominantly type B or C.

The isolated directions from the two sites have a mean declination of 76.9° and an inclination of 42.2° with an α_{95} of 7.5° and a k of 1123. Bedding correction gave no significant change in k or α_{95}.

Rock magnetism. Low field susceptibility showed a decrease at low temperature in the vicinity of 150°C followed by major increase above 350°C. Acquisition and demagnetization of IRM indicates the presence of two phases. There is contribution from a fine magnetite like carrier and also from a harder phase.

Interpretation. The fold test is indeterminate. The NRM appears to be carried by a combination of fine magnetite and a second phases which unblocks at low temperature. However, both phases carry a similar direction. The ratio of NRM:IRMs suggests a secondary magnetization.

3.2.1.5. The Permian Bacuit Formation from North Palawan Island. Sites PAL 90-5, -6, -7, -8 at which the Bacuit Formation were sampled on the mainland of North Palawan are shown in Fig. 9 and the results given in Table 1 and Fig. 12.

NRM. The NRM was very stable against AF demagnetization yielding only type D Zijderveld plots. However, thermal demagnetization unblocked the magnetization close to 300°C with type A plots. The results were indistinguishable from the present field direction before tectonic correction.

Rock magnetism. The acquisition and demagnetization of IRM revealed a very hard magnetic carrier. The combination of this result with the apparent Curie point in the vicinity of 300°C suggested by the unblocking temperature and by the weak field susceptibility result may reflect the presence of fine grain pyrrhotite. The magnetization does not respond to AF demagnetization, but the ratio of NRM:IRMs is in the range of secondary chemical magnetizations.

Interpretation. Because this magnetization is indistinguishable from the present field and the NRM:IRMs is too large for a primary DRM, we interpret this magnetization is a secondary magnetization acquired in the present field and do not include it in our analysis.

3.2.1.6. Summary of North Palawan Block results. Two sites in the Manguao volcanics give a direction with declination of 11.1° and inclination of 14.4° which is distinguishable from the present GAD field, but within the range of expected secular variation. We interpret this as a primary magnetization. The mean direction from sites in the Jurassic Busuanga chert and from the Cretaceous Guinlo in Busuanga Island are statistically indistinguishable and both sites fail a regional attitude test. These 11 are shown in Fig. 13, which reveals consistent inclinations, which are statistically indistinguishable at the 95% significance level. However, the

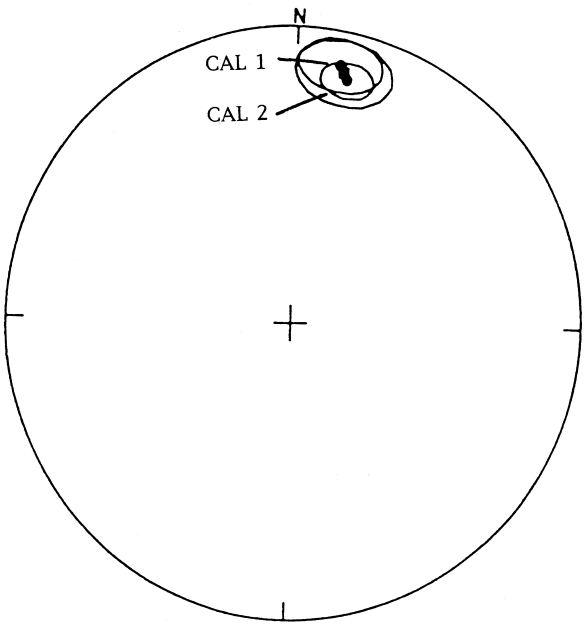


Fig. 10. Results from sites CAL 1 and 2 in the Manguao Volcanics of North Palawan.

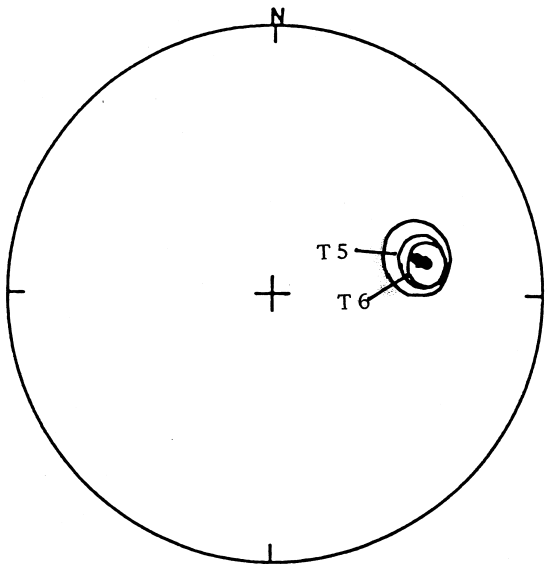


Fig. 11. Results from sites T5 and T6 in the Guinlo formation in Northern Palawan. The data are plotted uncorrected in situ. Correcting for bedding gives no change in k or α_{95}.

declinations spread over a range of some 40°. We interpret this as a secondary magnetization of unknown, but not recent age. The consistency of the inclinations suggests that the paleolatitude of these sites can be determined relatively reliably. In contrast, the spread in declination suggests that they have suffered differential rotation about a vertical axis since magnetization. The oldest formation sampled, the Permian Bacuit, gives a direction close to the present field before tectonic correction and an NRM:IRMs ratio too large to be a DRM. We interpreted this as a present field remagnetization.

3.2.2. South Palawan Block

The South Palawan Block is dominated by ophiolites. In the northernmost part of the SPB the Palawan ophiolite is sheared and faulted in the region of the boundary between the two blocks. Further south the Calatuigas ophiolite is less deformed. We collected from the basalts of each ophiolite. The basalts from both ophiolites are named the Espina basalts. The Palawan ophiolite is thought to be Middle Eocene from K–Ar dating (Raschka et al., 1985). However, paleontological evidence gives an older age of Cretaceous (Casasola, 1956), which would be similar to the ophiolites in North Sabah. The Mt Beaufort Ultramafics is the dominant unit and is overlain by the Stavely Range Gabbro. The thrusting of the ophiolite over the quartzofelspathic turbidites of the Panas formation has yielded a variety of low grade metamorphic rocks.

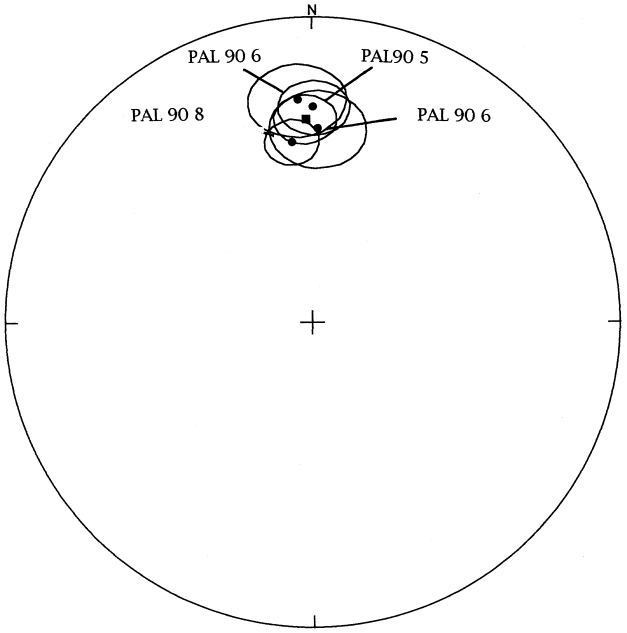


Fig. 12. Results from Bacuit Formation in North Palawan Island. The data are plotted uncorrected in situ.

3.2.2.1. Paleomagnetic results from Basalts of Palawan Ophiolite. Sites BAC 1, 2, 7, and 11, IRA 1 and 2, and P86 3 and 5 were drilled in the flows of the Palawan ophiolite (Fig. 14).

NRM. These samples responded well to AF demagnetization yielding primarily type A or B Zijderveld plots. The within site scatter was small with α_{95} in the vicinity of 10°, but the between site scatter was extreme, whether the data were plotted uncorrected, or in stratigraphic coordinates.

Rock magnetism. Low field susceptibility yielded Curie points consistent with magnetite as the predominant carrier at sites BAC 1 and 11. and AF demagnetization characteristics were consistent with the presence of fine magnetite.

Interpretation. The combination of low within site scatter, satisfactory demagnetization characteristics, but large between site scatter suggests that sites have experienced differential rotations during localized deformation after acquiring their magnetization. This is consistent with the relatively small size of the thrust sheets in which they are located and the strong deformation related to the proposed north south faulting at the boundary between the North and South Palwan Blocks in this region. Thus these results do not provide useful information for regional tectonics, although more detailed studies might help to unravel patterns of local deformation in these thrust sheets.

3.2.3.1. Paleomagnetism of the Calatuigas Ophiolite. Sites from Southern Palawan were drilled in the basalts and andesites of the Calatuigas Ophiolite. The results from sites in the Espina Basalts (Fig. 3) of the Calatuigas Ophiolite from about 10 miles east of Quezon (A 1, 2, and Pan 3 in Fig. 15a) and in the Brookes

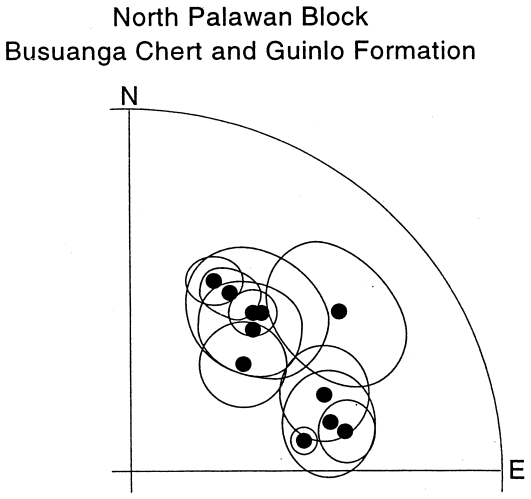


Fig. 13. Combined results from Busuanga Chert and Guinlo Formation of Busuanga Island (North Palawan Block). The data are plotted uncorrected in situ.

Point region (B 4, 7, 10, and 12 in Fig. 15b) are presented together (Fig. 16 and Table 2).

NRM. Both alternating-field and thermal demagnetization were effective in eliminating a very soft low blocking temperature magnetization in the basaltic rocks (B 4, 7, A1, 2, Pan 3) with almost all Zijderveld diagrams being type A or B. Similar results were obtained from the andesites from B10. Most of the specimens were demagnetized by the AF method. Duplicate specimens from some cores were thermally demagnetized and gave the same direction as that obtained from AF demagnetization.

Plots of the in situ mean directions from individual sites exhibit a NW–SE orientation (Fig. 16a), which on correction for bedding yielded approximately antipodal low inclination directions (Fig. 16b). To determine the in situ geographic and corrected stratigraphic mean direction of the sites, the results from the southeasterly sites were reversed. This gave an in situ mean declination of 293.3° and inclination of 16.5° with α_{95} 23.9° and k of 7.3 and a corrected mean declination of 296.2° and inclination of −0.4° with α_{95} 17.1° and k of 13.5. The directions thus technically pass a regional attitude test. However, this is an example of an unstable attitude test. Elimination of the single, site B 4, which has realtively poor statistics gives a failed fold test. Elimination of a second site B 7 gives a positive field test with in situ declination 294.5°, inclination 15.4, α_{95} 14.6 and k 28.5 with corrected stratigraphic results declination 293.9, inclination 5.8, α_{95} 12.6 and k 37.7.

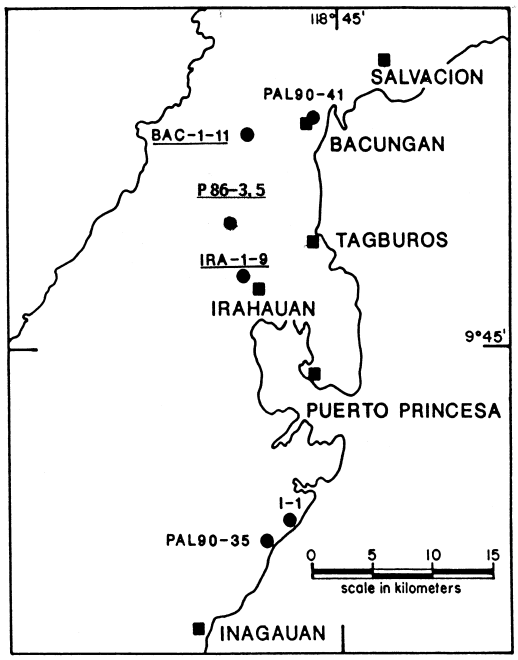


Fig. 14. Site location map for Central Palawan.

Rock magnetism. Observations of weak field susceptibility as a function of temperature and the thermal demagnetization results are consistent with a low titanium titanomagnetite carrying the NRM in all of these basalts and andesites. There are variations in grain size indicated by AF characteristics and weak field susceptibility with B 7 and A 2 having more coarse grained multidomain magnetite while the andesite of B 10 and the basalt of B 4 are finer. The NRM:IRMs plots of A1, A2, Pan 3, B 10 and 12 are typical of TRM acquired in a weak magnetic field, whereas B4 and 7 show lower ratios of NRM:IRMs indicating that the NRM cannot be a TRM.

Interpretation. The marginal attitude test does not give confidence in a simple comparison between k values of the whole population as a means of interpreting the magnetism of these flows in the Calatuigas Ophiolite. However, the demagnetization characteristics and the ratio of NRM:IRMs suggest that the samples from sites A1 and 2, Pan 3, and B10 and 12 carry primary TRM, whereas sites B 4 and 7 do not. We therefore use only those sites which have magnetization consistent with a primary TRM acquired when they were formed on the seafloor. This yields a positive attitude test for the five sites and a declination of 294.5°, an inclination of 15.4° with an α_{95} 14.6° and a k of 28.6 prior to tectonic correction and 293.9, 5.8, 12.6 and 37.7 respectively after tectonic correction. This test remains positive upon exclusion of any of the sites. The alteration of B 4 may have taken place on the sea floor before obduction because its tectonically corrected value has a smaller α_{95} and a higher k than the in situ value, but the site is not included in the final analysis.

3.2.3.2. Paleomagnetic results from the sedimentary rocks of the SPB. Numerous sites were drilled in the Panas and Pandian Formations. The magnetizations were so weak and scattered that only sites 9015, 9019 and 9020 (Fig. 17) yielded results which permitted site means to be calculated. The wide scatter of the site means make these results uninterpretable. The shallow northwesterly direction of 9015 and 9020 are somewhat similar to the results from the flows of the Calatuigas ophiolite, but given the poor nature of the data, it seems unwise to interpret these data further.

3.2.3.3. Summary of Paleomagnetic results from South Palawan. The flows of the Calatuiges ophiolite are the only units to give useful results in the South Palawan Block. They give antipodal shallow directions in a northwesterly and southeasterly directions. Assuming that the northeasterly direction is the normal direction, which is to assume the flows were formed in the northern hemisphere, these results indicate that the flows have been rotated 65° CCW with respect to the Geo-

centric Axial Dipole (GAD) field and translated northward from a latitude of 2.5°N ± 0.6° to their present latitude of 9°N. With data from these low latitudes, we cannot escape the polarity indeterminancy, so that the site of origin could have been in the southern hemisphere at latitude of 2.5°S and the rotation 115° clockwise. Unfortunately there are no reliable younger results from the SPB to resolve the indeterminacy through establishing the progression of rotations as a function of age.

3.3. Summary of Paleomagnetic results

While the paleomagnetic data from Palawan did not yield as many reliable results as might have been expected given the magnitude of the initial collection,

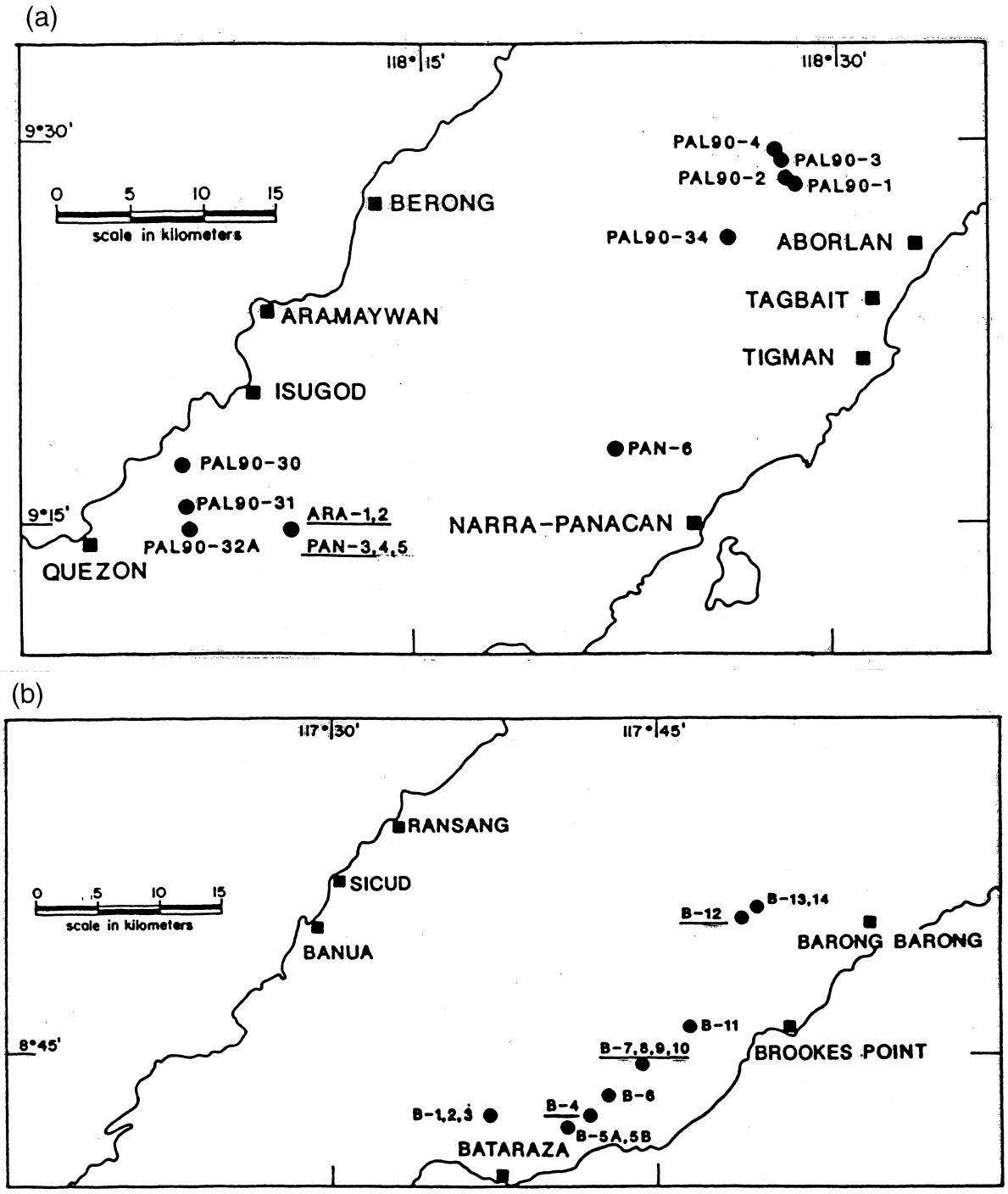


Fig. 15. Site location map for Calatuigas ophiolite; (a) Quezon Region and (b) Brooke's Point region.

they do clearly reveal that the regions sampled in the North and South Palawan blocks have two very different paleomagnetic signatures. Fig. 18 gives the declination of all of the results discussed above, with the width of the pie slice giving the magnitude α_{95}. The upper plot in each pair gives the in situ result before tectonic correction and the lower the stratigraphic or tectonically corrected result.

Results from Busuanga Island show CW declinations, which are more consistent before tectonic correction than in stratigraphic coordinates. In the northernmost part of Palawan Island there are results similar to the present field direction in the young flows (CAL1 and 2) and in the oldest Bacuit formation (PAL 90 5,6 and 8) and strongly CW declinations in the Cretaceous Guinlo formation similar to the results from Busuanga Island. Thus the dominant paleomagnetic signature of the NPB is CW declination with inclinations indicating magnetization in more northerly latitudes (Table 1).

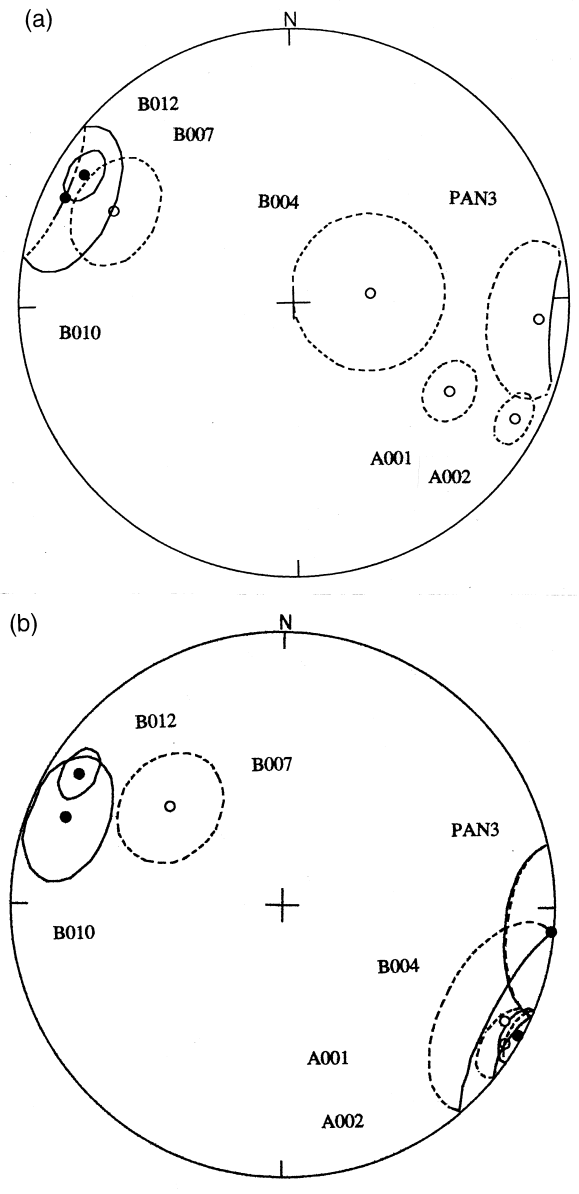


Fig. 16. Results of Sites in flows of Calatuigas Ophiolite. (a) Site means plotted in uncorrected in situ coordinates. (b) Site means plotted in corrected stratigraphic coordinates.

In the narrowest part of the island near Puerto Princessa, the directions from the Palawan Ophiolite are strongly scattered (IRA 1 and 6, BAC 1, 2, 7, and 11, and P86 3 and 5), but further south the results from the Calatuigas Ophiolite (A1 and 2, Pan 3, and B10, and 12) are more internally consistent showing CCW declinations and have demagnetization characteristics and intensities of NRM consistent with primary TRM. In the sediments in the southernmost part of the island there are some directions similar to those in the ophiolites (PAL 90 20 and 15), but others such PAL 90 19 are inconsistent, so that without further work these results from the sediments are not interpretable. The paleomagnetic signature of the SPB (Table 2) comes from the Calatuigas Ophiolite and indicates CCW rotation with possible origins to the south of the present latitude.

4. Discussion

In the following, we discuss the implications of the results for tectonic models of (1) the South China Sea (2) the Sulu and Celebes Seas and (3) Philippine Archipelago. We also note in passing that although we were able to get coherent and interpretable patterns of magnetization from formations in the North and South Palawan blocks, at the boundary of the two regions no coherent pattern emerged, which draws attention to possible pitfalls in carrying out paleomagnetism in regions of strong deformation.

4.1. South China Sea

The paleomagnetic results from the NPB suggest southward drift and clockwise (CW) rotation. The key results are those from (1) the Guinlo and the Busuanga Chert, from Busuanga island and (2) the Guinlo on the main island of Palawan. The inclinations are internally consistent giving paleolatitudes of 25.9°, 26.4° and 24.3° N, with a mean of 25.5°, and standard deviation of 1.1°. This places the North Palawan Block at the time of magnetization of these sediments at the latitude of the southern margin of China during the Late Cretaceous, e.g. Holloway (1982), Hall (1996), and Richter et al. (1999). The results also require CW rotation with respect to South China. Moreover, because the VGPs from Busuanga Island and from Northern Palawan indicate differential CW rotation,

Table 2
Paleomagnetic results from the South Palawan Block

Site No.	Age	Formation	Lithology	Latitude	Longitude	N	Geographic mean					Stratigraphic mean			
							Declination (deg)	Inclination (deg)	α_{95} (deg)	k	Bedding correction	Declination (deg)	Inclination (deg)	α_{95} (deg)	k
BAC-1	Eocene	Palawan O	Basalt	9.9N	118.7E	7	333	−0.4	12.7	23.6	290 66NE	315.2	−33.6	16.9	13.8
BAC-2	Eocene	Palawan O	Pillow basalt	9.9N	118.7E	8	312.6	33.1	9.4	35.9	290 66NE	330.6	−1.0	9.5	34.9
BAC-7	Eocene	Palawan O	Pillow basalt	9.9N	118.7E	5	14.9	30.5	14.3	29.4	290 66NE	14.6	−35.2	16	34
BAC-11	Eocene	Palawan O	Basalt	9.9N	118.7E	9	85.1	−67.2	15.1	11.2	290 66 NE	179.2	−30.7	15.5	11.9
IRA-1	Eocene	Palawan O	Pillow basalt	9.8N	118.7E	9	53.8	16.2	11.4	21.5	0	53.8	16.2	11.4	21.5
IRA-6	Eocene	Palawan O	Basalt	9.8N	118.7E	10	278.6	−41.4	7.8	39	0	278.6	−41.4	7.8	39
P86.3	Eocene	Palawan O	Basalt	9.8N	118.7E	7	207.9	−53.2	10.9	31.4	290 66NE	205.1	12.6	10.9	31.5
P86.5	Eocene	Palawan O	Basalt	9.8N	118.7E	6	255.4	−55.9	11.5	34.9	290 66NE	227.5	−2.6	11.5	34.7
PAL90-15	Eocene	Panas	Sandstone	8.7N	117.5E	5	308.1	−11.2	11.2	47.3	270 30N	292.9	−24.8	11.3	47.1
PAL90-19	Eocene	Panas	Sandstone	8.5N	117.3E	7	50.3	−23.5	12.8	23.3	7 77SE	347.8	−44	12.9	23
PAL90-20	Eocene	Panas	Sandstone	8.7N	117.5E	4	278.1	−15.8	24.6	15	274 37NE	267.7	−14.4	24.6	14.9
A-1	Cretaceous	Calatuigas O	Pillow basalt	9.2N	118.2E	9	120.7	34.4	8.4	38.2	215 30W	121.6	−4.4	8.4	38.7
A-2	Cretaceous	Calatuigas O	Pillow basalt	9.2N	118.2E	12	119	−9.9	6.3	48.6	185 12W	118.6	1.1	6.3	48.5
PAN-3	Cretaceous	Calatuigas O	Basalt	9.2N	118.2E	5	95.2	−11.6	18.6	17.9	185 12NW	95.2	0.4	18.6	17.9
B-4	Cretaceous	Calatuigas O	Basalt	8.7N	117.7E	6	84.1	−67	27.9	6.7	Variable	117.1	−9.73	22.5	9.7
B-7	Cretaceous	Calatuigas O	Basalt	8.7N	117.7E	8	298.2	−27	14.5	15.5	170 20W	310.4	−43.7	16.2	12.7
B-10	Cretaceous	Calatuigas O	Andesite	8.7N	117.7E	12	302.4	11.3	6.4	47.1	0	302.4	11.3	6.4	47.1
B-12	Cretaceous	Calatuigas O	Pillow basalt	8.8N	117.8E	10	115.1	−8.5	18.7	13.8	280 25SW	116	2	20.5	11.6
		Mean				7	293.3	16.5	23.9	7.3	Variable	296.2	−0.4	17.1	13.5
		Mean*				5	294.5	15.4	14.6	28.5	Variable	293.9	5.8	12.6	37.7

we can be sure that at least some of this rotation is local and not plate wide (Fig. 19). These results are similar to those obtained from Mindoro by McCabe et al. (1987), although those results represent a pre-folding direction, while the results reported here fail an attitude test and probably reflect the profound Cretaceous remagnetization event observed in South China (Kent et al., 1987; Dobson and Heller, 1992) and S.E. Asia (Chen and Courtillot, 1989). The absence of results from sites in the SPB below the ophiolites precludes comparison of the location of the basement of the SPB paleomagnetically.

4.2. Celebes and Sulu Sea

The magnetization of the Espina Basalts of the Calatuigas ophiolite is distinct from the results from the North Palawan Block and appears to be a primary magnetization. The Espina basalt data require 65° of CCW rotation with respect to GAD and give a paleolatitude of 2.5°N. This places the sea floor at the time of magnetization at the present latitude of the Celebes Sea. The paleomagnetic results obtained by Shibuya et al. (1991) during Leg 124 of the Ocean Drilling Project in the Celebes and Sulu Seas were from Eocene and younger basaltic basement of the Celebes Sea (Silver and Rangin, 1991), which is younger than the age of the basalt in Palawan. Thus while the Espina basalt of South Palawan cannot be Celebes Sea crust, it appears to have taken part in a similar rotation and to have come from equatorial latitudes. The paleomagnetic results also permit a paleolatitude of 2.5° S and a larger CW rotation because of the equatorial indeterminacy of polarity by declination alone. However, we prefer the interpretation of the smaller rotation which is comparable with that observed by Shibuya et al. (1991) in the Celebes Sea. No matter which interpretation is taken, this result demonstrates that the Calatuigas ophiolite of South Palawan is not obducted South China Sea floor. The paleomagnetic data are consistent with the view that the ophiolite was thrust from the south over the underlying sequences (De los Santos, 1959). Hinz and Schluter (1985) also suggested this scenario. The Calatuigas ophiolite and probably

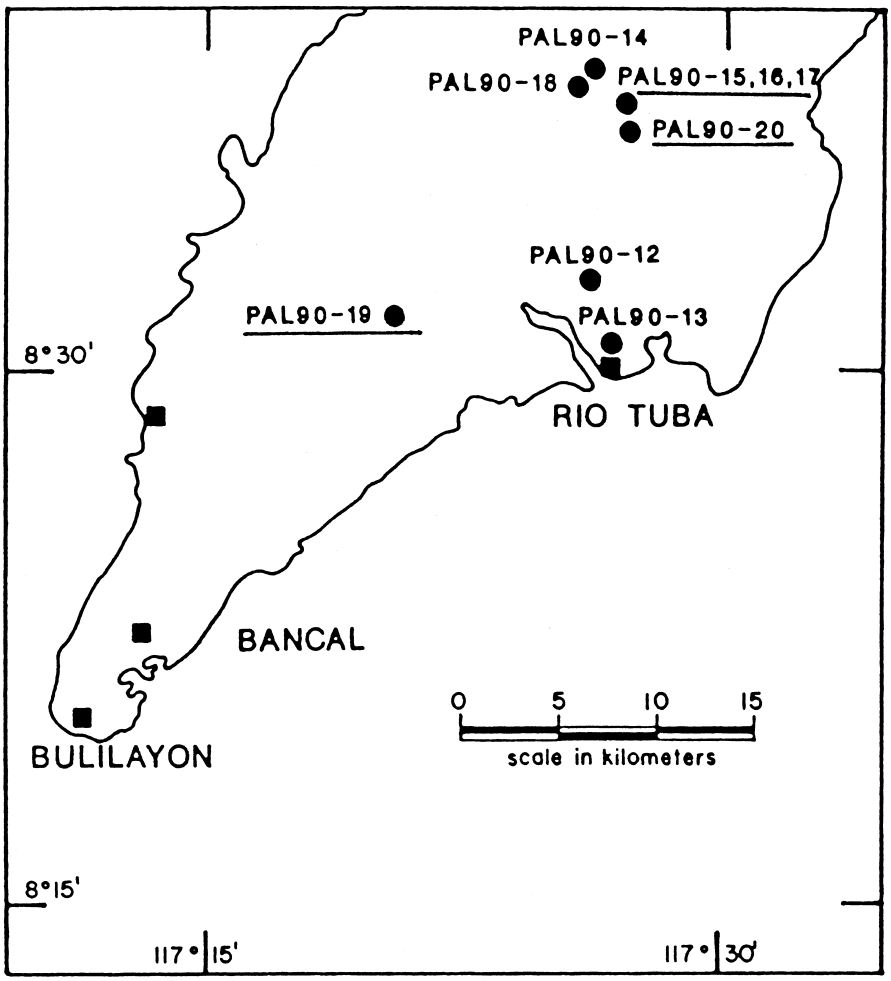


Fig. 17. Site locations in sediments of southernmost Palawan.

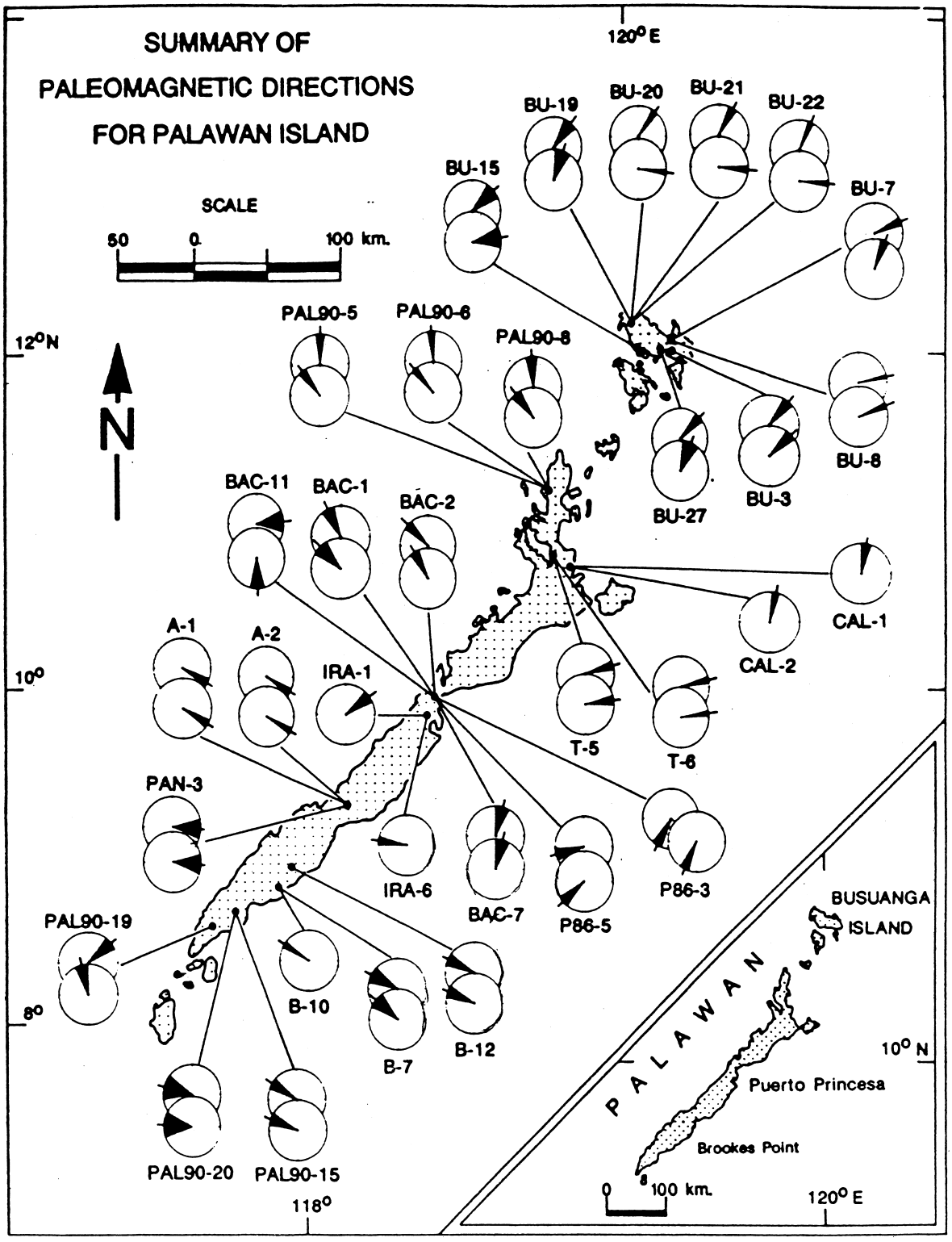


Fig. 18. Summary of paleomagnetic results from Palawan. The upper circle of each pair gives the declination in situ and the lower corrected to stratigraphic coordinates. The pie slice indicates the intersection of the α_{95} cone with the horizontal plane. (1) The results from Busuanga Island (Bu) are from the Jurassic Busuanga Chert and the Cretaceous Guinlo Formation. They fail a regional attitude test and the upper circles indicate consistently clockwise rotated declinations in situ. (2) The youngest results from the north of the main island are from the Pleistocene Manguao Volcanics (CAL-1 and 2) and yield directions close to the present field. In the Guinlo (T-5 and T-6), the directions are similar to those in Busuanga Island, but are more strongly CW rotated. The Permian Bacuit formation sites (PAL90-5,6 and 8) are remagnetized in the present field direction as shown in the upper circles. (3) In Central Palawan the results come from the ophiolites in a region which is tectonically disturbed and the scatter in directions is extreme. (4) In Southern Palawan the results from the sediments (PAL-15,19 and 20) are scattered and the α_{95} is large, but the remainder of the sites, which are from the Espina ophiolite, consistently give NE/SW shallow directions, reflecting CCW rotation.

similar ophiolites in Northern Sabah appear to have originated as Cretaceous sea floor created to the south of their present latitudes and experienced CCW rotations and northward migration.

4.3. Philippine archipelago

The similarity of the results from the North Palawan Block with those from Mindoro (McCabe et al., 1987) confirm the conventional view that these two regions were part of a landmass brought southward by the opening of the South China Sea (e.g. Holloway, 1982). Moreover, at some point this landmass must have collided with the remainder of the Philippine Archipelago. The similarity of the paleomagnetic results from the Espina Basalts in the South Palawan Block and the Celebes Sea data reported by Shibuya et al. (1991) suggest that both regions experienced CCW rotations of some 60° during the Eocene and Oligocene. The existence of directions of magnetization in Cretaceous sediments from Cebu indicating about 60° of CCW rotation (McCabe et al., 1987) raises the possibility that a substantial region including the Celebes Sea and parts of the Philippine archipelago may have experienced this Eocene–Oligocene rotation. Indeed, they may be part of a far larger region stretching from the Philippines to Peninsular Malaysia and including Borneo, which experienced this CCW rotation in early Tertiary times.

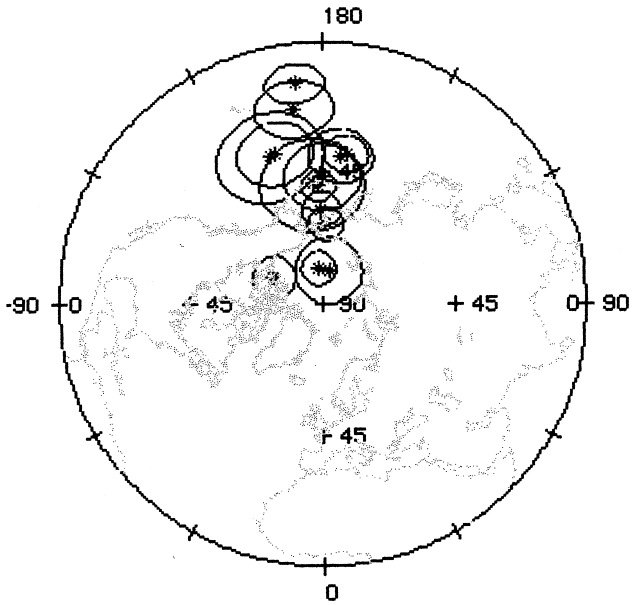


Fig. 19. Comparison of poles from the North Palawan Block with South China poles. The three high latitude poles are Late Cretaceous poles from China (Enkin et al., 1991). The remaining poles are from the North Palawan Block. The distribution of these Palawan poles is consistent with magnetization at the latitude of southernmost China during the Cretaceous, southerly motion and clockwise rotation with respect to South China. The elongation of the distribution of the Palawan VGPs along a small circle about the site suggests that the various sites have suffered different amounts of clockwise rotation about vertical axes.

5. Conclusion

The paleomagnetic results from Palawan are disappointing and conclusions are correspondingly weak. The results from the North Palawan Block are, however, not inconsistent with the conventional tectonic history of an origin on the southern border of China followed by southerly displacement and clockwise rotation during South China Sea opening. The results from the Calatuigas ophiolite of the South Palawan Block suggest northerly displacement, obduction from the south, and CCW rotation of 60° with respect to the geocentric axial dipole (GAD) field.

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